Design a Personal Portfolio Page Using HTML and CSS

1. Objectives

- To design and develop a fully static personal portfolio web page using only HTML5 and CSS3, without any external libraries or frameworks.
- To apply core CSS concepts — including Flexbox, CSS Grid, animations, transitions, media queries, and CSS variables — to build a responsive and visually appealing layout.
- To present personal information such as skills, projects, education, experience, and contact details in a structured, professional, and user-friendly manner.

2. Software / Hardware Requirements

Software Requirements:

- Operating System: Windows 10/11, macOS, or any Linux distribution
- Code Editor: Visual Studio Code, Sublime Text, or Notepad++
- Web Browser: Google Chrome, Mozilla Firefox, or Microsoft Edge (latest version)
- Languages Used: HTML5 and CSS3 (no JavaScript frameworks or external libraries required)
- Google Fonts API: Used via CDN link tag to load custom typefaces (Syne, DM Sans)
- Server: Not required — the HTML file opens directly in any browser as a local file

Hardware Requirements:

- Processor: Intel Core i3 or equivalent (any modern CPU)
- RAM: Minimum 2 GB (4 GB recommended for smooth browser performance)
- Storage: At least 10 MB free disk space
- Display: Minimum resolution of 1024 x 768 (1920 x 1080 recommended)

- Internet Connection: Required only to load Google Fonts; the page itself is fully offline-compatible

3. Theory

3.1 Introduction to HTML and CSS

HTML (HyperText Markup Language) is the skeleton of every web page — it defines the structure and content using elements represented by tags. CSS (Cascading Style Sheets) is the skin and clothing — it controls the visual presentation, layout, and responsiveness of that structure.

Together, HTML and CSS form the foundational duo of front-end web development. A personal portfolio page is one of the most practical and common projects used to learn these technologies, because it requires the developer to apply a wide range of HTML elements and CSS properties in a real, meaningful context.

3.2 What is a Personal Portfolio Page?

A personal portfolio page is a static website that serves as a digital resume and showcase for a developer, designer, or student. It introduces who the person is, highlights their technical skills, lists their projects and work experience, and provides a way to get in contact. Unlike a traditional paper resume, a portfolio page is interactive, visually rich, and accessible globally via a browser.

Key characteristics of a good portfolio page include: a clean and professional visual design, easy navigation, mobile responsiveness, fast loading speed (since it is static), and clear sectioning of information.

3.3 Structure of the Portfolio Page

The portfolio designed in this assignment is a single-page application (SPA-like) with smooth in-page navigation. It is divided into the following major sections:

Section	Content	Key CSS Used
Navigation Bar	Fixed top bar with logo and smooth-scroll	position: fixed; backdrop-filter: blur()

Section	Content	Key CSS Used
	links	
Hero Section	Full-screen intro with name, role, bio, CTA buttons, and stats	min-height: 100vh; display: flex; animation
Skills Section	Grid of skill cards grouped by category (languages, tools, etc.)	display: grid; grid-template-columns; hover transitions
Experience Section	Card showing internship role, company, dates, and responsibilities	border-left accent; flex layout; list styling
Projects Section	Cards for each project with title, description, and tech stack tags	display: grid; box-shadow; transform on hover
Education Section	Cards for B.Tech and HSC qualifications with score badge	grid layout; color-coded badge; border hover
Achievements Section	Icon + text cards for LeetCode, RAG system, interests	grid; emoji icons; subtle hover lift
Contact Section	Centered card with email, LinkedIn, and phone links	text-align: center; flex; gradient top border
Footer	Simple credit line at the bottom	border-top; text-align: center; padding

3.4 Key HTML Elements Used

- <nav>: Semantic navigation bar element holding the logo and navigation links.
- <section>: Used to divide the page into logical, semantic sections (hero, skills, projects, etc.).
- <h1>, <h2>, <p>: Heading and paragraph tags for textual hierarchy and content.
- <a>: Anchor tags used for internal smooth-scroll navigation and external links (LinkedIn, email, phone).
- / : Used to render bulleted lists of experience responsibilities.
- and : Generic container elements used for grouping and styling sub-components such as cards, badges, and tags.
- <footer>: Semantic footer element for the credit line at the bottom of the page.

3.5 Key CSS Concepts Used

The following table summarises the major CSS properties and concepts applied in this portfolio:

CSS Property / Concept	Purpose	Usage in Portfolio
font-family	Sets the typeface/font for an element	font-family: 'Syne', Arial, sans-serif
font-size	Defines the size of text	font-size: 16px / clamp(14px, 2vw, 18px)
font-weight	Sets the thickness of text (bold/normal)	font-weight: 700 / bold / 300
color	Sets the foreground (text) color	color: #e2e8f0 / rgba(0,0,0,0.8)
background	Sets background color or gradient	background: #080c10 / linear-gradient(...)
padding	Inner spacing between content and border	padding: 20px / padding: 10px 40px
margin	Outer spacing outside the border	margin: 0 auto / margin-top: 20px
display	Defines the display type of an element	display: flex / grid / block / none
flexbox	Layout method for 1D arrangement	display: flex; gap: 16px; flex-wrap: wrap
grid	Layout method for 2D arrangement	display: grid; grid-template-columns: repeat(...)
border	Adds border to an element	border: 1px solid #1e2d40
border-radius	Rounds the corners of an element	border-radius: 8px / 50% / 100px
box-shadow	Adds shadow around an element	box-shadow: 0 12px 40px rgba(0,0,0,0.3)
text-decoration	Decorates text (underline, none, etc.)	text-decoration: none / underline
position	Controls positioning method	position: fixed / relative / absolute
transition	Animates property changes smoothly	transition: all 0.2s ease
@keyframes	Defines CSS animation keyframes	@keyframes fadeUp { from {...} to {...} }
animation	Applies keyframe animation to element	animation: fadeUp 0.6s 0.2s ease both
media query	Applies styles at specific breakpoints	@media (max-width: 768px) { ... }
backdrop-filter	Applies filter to background area	backdrop-filter: blur(12px)

3.6 CSS Variables and Design System

The portfolio uses CSS custom properties (variables) defined in the :root selector to establish a consistent design system. Variables such as --bg, --accent, --card, --text, and --border are reused

throughout the stylesheet, making the design easy to maintain and theme. This mirrors the approach used in professional design systems and component libraries.

3.7 Responsiveness

Media queries are used to adapt the layout for smaller screen sizes (max-width: 768px). On mobile, the navigation links are hidden, horizontal padding is reduced, and grid columns collapse to a single-column layout. This ensures the portfolio is usable across all device sizes — desktop, tablet, and smartphone.

3.8 Animations and Transitions

CSS @keyframes animations (fadeUp) are applied to hero section elements with staggered animation-delay values, creating a polished entrance effect when the page loads. Hover transitions using the CSS transition property are applied to cards, buttons, and links, giving the page a responsive, interactive feel without any JavaScript.

4. Conclusion / Analysis

This assignment successfully demonstrated the design and development of a complete personal portfolio web page using only HTML5 and CSS3. The project covered a wide spectrum of front-end concepts — from semantic HTML structure to advanced CSS techniques including Flexbox, Grid, custom properties, keyframe animations, and responsive design with media queries.

A significant learning outcome was understanding the difference between structural HTML (semantic tags like <nav>, <section>, <footer>) and presentational CSS. Separating concerns between structure and style results in code that is more readable, maintainable, and scalable.

The use of CSS variables proved especially valuable — defining a central color palette and reusing it throughout the stylesheet meant that the entire visual theme could be changed by modifying just a few lines of code. This reflects the real-world practice of building design tokens.

The project also reinforced the importance of responsive design. By testing the layout at different viewport widths and applying appropriate media queries, the portfolio was made accessible across

devices. In today's multi-device browsing environment, responsiveness is not optional — it is a core requirement.

Overall, this exercise provided practical, hands-on experience in building a production-quality static web page and highlighted how HTML and CSS, even without JavaScript, are powerful enough to create professional, animated, and fully responsive user interfaces.